Thermodynamic and Kinetic Considerations of Synthesis Reaction Process for Metal Nanorods/Nanowires by Solution Combustion Synthesis

Abstract

Solution combustion synthesis (SCS) is a rapid, self-sustained exothermic technique for producing metal and metal oxide nanopowders. This paper presents a detailed thermodynamic and kinetic analysis of the SCS process, with specific focus on copper-based systems as model materials for understanding metal nanostructure formation. The thermodynamic framework based on Gibbs free energy minimization is established, demonstrating that adiabatic flame temperature (T_ad) increases with fuel-to-oxidizer ratio (φ) up to a maximum of approximately 2480 K at φ = 3 for $Ni(NO_3)_2$-glycine systems . Kinetic analysis using the Merzhanov-Khaikin method reveals that fuel-rich compositions exhibit lower activation energies for metal formation compared to fuel-lean compositions producing oxides . Experimental evidence from quenched combustion waves confirms a two-stage reaction pathway: oxide phase formation in the combustion front followed by reduction to metallic copper in the post-combustion zone . The fundamental thermodynamic and kinetic constraints that prevent SCS from achieving one-dimensional metallic nanostructure growth are analyzed, explaining why solution-phase methods employing screw dislocation-driven growth remain the predominant route to copper nanowires .

Keywords: Solution combustion synthesis; Thermodynamics; Kinetics; Adiabatic temperature; Activation energy; Copper nanoparticles; Reaction pathway

1. Introduction

Solution combustion synthesis (SCS) represents a unique class of self-propagating high-temperature synthesis (SHS) conducted in aqueous or gel media. The technique involves an exothermic redox reaction between metal nitrates (oxidizers) and organic fuels (reducing agents), initiated at relatively low temperature and driven by internal chemical energy . Understanding the thermodynamics and kinetics of SCS reactions is essential for rational design of synthesis processes, particularly for controlling product phase (metal versus oxide) and morphology.

The thermodynamic analysis of SCS centers on calculating the adiabatic flame temperature (T_ad) and equilibrium product composition using the Gibbs free energy minimization principle . This provides insight into the maximum temperature achievable and the thermodynamically favored products under adiabatic conditions. Kinetic analysis, on the other hand, addresses the reaction mechanisms, rate-limiting steps, and activation energies governing combustion wave propagation and product formation .

This paper presents a comprehensive thermodynamic and kinetic analysis of SCS for metal nanostructure synthesis, using copper as the model system. The theoretical framework is established, experimental validation is presented, and the implications for one-dimensional growth are discussed.

2. Thermodynamic Framework

2.1 Gibbs Free Energy Minimization

The equilibrium state of an SCS reaction system at a given temperature and pressure is determined by minimizing the Gibbs free energy of the system. For a multicomponent, multiphase system containing gaseous and condensed species, the total Gibbs free energy is expressed as :

$$
G_{total} = \sum_i n_i \mu_i
$$

where n_i is the number of moles of species i and μ_i is its chemical potential. At equilibrium, the system composition and temperature are determined by simultaneous solution of the Gibbs energy minimization and enthalpy balance equations under adiabatic conditions.

Software packages such as THERMO and ASTRA-4 have been developed for these calculations. These programs contain thermodynamic databases with temperature-dependent heat capacities for gaseous and condensed species, enabling accurate computation of equilibrium compositions at the calculated adiabatic temperature.

2.2 Adiabatic Flame Temperature

The adiabatic flame temperature (T_ad) represents the maximum temperature achievable under adiabatic conditions and is a critical parameter governing product phase, crystallinity, and particle size. T_ad is calculated by solving the enthalpy balance equation :

$$
\sum \Delta H^0(T_0) = \sum \Delta H^0_{products}(T_{ad})
$$

or in simplified form using mean heat capacity:

$$
T_{ad} = T_0 + \frac{\Delta H^0}{C_p}
$$

where ΔH^0 is the reaction enthalpy at initial temperature T_0, and C_p is the mean heat capacity of products.

For the $Ni(NO_3)_2$-glycine system, thermodynamic calculations reveal that T_ad increases with φ, reaching a maximum of **2480 K** at φ = 3 with zero water content . The effect of water content is significant: increasing water reduces T_ad due to the enthalpy of vaporization . This has practical implications, as precursor drying time controls residual water content and thus combustion temperature.

2.3 Effect of Fuel-to-Oxidizer Ratio (φ)

The fuel-to-oxidizer ratio φ is defined such that φ = 1 corresponds to stoichiometric oxygen concentration (no atmospheric oxygen required for complete fuel oxidation), φ > 1 indicates fuel-rich conditions, and φ < 1 indicates fuel-lean conditions .

Thermodynamic analysis of $Cu(NO_3)_2$-fuel systems reveals the following trends:

Table 1: Effect of φ on Thermodynamic Parameters and Product Phase

φ Condition	T_ad Trend	Dominant Gas Species	Product Phase
φ < 1 (lean)	Lower T_ad	NO, O_2	CuO
φ = 1 (stoichiometric)	Moderate T_ad	CO_2, H_2O, N_2	CuO + Cu_2O
φ > 1 (rich)	Higher T_ad	H_2, CO, NH_3	Metallic Cu

At φ > 1, the excess fuel generates a **hydrogen-rich reducing atmosphere** in the reaction front, which is the necessary condition for metallic phase formation . The protective atmosphere formed from fuel oxidation products enables synthesis of pure metal in common atmosphere .

2.4 Effect of Fuel Type

Different fuels exhibit distinct thermodynamic and combustion characteristics due to variations in energy density and decomposition pathways :

- **Glycine ($C_2H_5NO_2$)**: High energy density, vigorous combustion with rapid temperature and pressure increase. Exhibits two distinct reaction phases: initial thermal runaway followed by explosive event .

- **Citric acid ($C_6H_8O_7$)**: Lower exothermicity with longer combustion period. Complex decomposition pathway involving aconitic, itaconic, and formic acids . Often transitions to smoldering mode.

- **Urea (CH_4N_2O)**: Produces hypergolic gas mixtures (HNCO, NH_3, NO_x) leading to foam formation. Maximum combustion temperature higher than glycine .

- **Hexamethylenetetramine (HMT, $C_6H_{12}N_4$)**: Enables one-step synthesis of pure metallic Cu nanoparticles under normal air atmosphere .

The shape of the time-temperature profile and maximum combustion temperature greatly depend on the fuel and the precursor decomposition process .

3. Kinetic Considerations

3.1 Merzhanov-Khaikin Method for Activation Energy

The activation energy (E) of SCS reactions can be determined using the Merzhanov-Khaikin approach, which correlates combustion wave velocity (V_c) and maximum combustion temperature (T_c) :

$$
\ln\left(\frac{V}{T_c}\right) = A - \frac{E}{2RT_c}
$$

where A represents all constant values in narrow reaction zones. The slope of the $\ln(V/T_c)$ versus $1/T_c$ plot allows determination of the activation energy. Multiple trials with different amounts of diluent are needed to generate combustion parameter datasets .

Application of this method to SCS systems requires special consideration. Unlike conventional combustion synthesis, the thermophysical parameters and densities of SCS samples change during reaction. Drastic local temperature oscillations occur due to rapid gas release and substantial porosity increase. Smoothing algorithms must be applied to temperature profiles before computing derivatives, which significantly influences the precision of measured activation energies .

3.2 Activation Energy Trends

Recent kinetic studies on $Ni(NO_3)_2$-fuel systems reveal significant trends in activation energies :

- **Fuel-lean solutions (producing NiO)**: Higher activation energy
- **Fuel-rich solutions (producing Ni)**: Lower activation energy

The reduction of activation energies with increasing φ is related to a **change in the rate-limiting stage**: from endothermic decomposition of individual reactants to exothermic decomposition of coordinate compounds formed between the reactants . This shift in mechanism explains the enhanced reactivity and metallic phase formation under fuel-rich conditions.

For copper systems, the activation energy for metallic Cu formation under fuel-rich conditions is expected to be similarly reduced compared to oxide-forming compositions, though specific values depend on the fuel type and φ ratio.

3.3 Reaction Rate and Combustion Mode

The combustion mode depends on fuel type and heating conditions :

- **Volume combustion**: Glycine and glucose exhibit rapid, narrow temperature peaks with high dT/dt and dP/dt values. Glycine shows two distinct phases: thermal runaway followed by explosive event .

- **Self-propagating mode**: Citric acid transitions to smoldering mode with slow exothermic reaction and longer combustion period .

The combustion mode directly affects the temperature-time history experienced by the product, which in turn influences crystallite size, phase purity, and morphology.

4. Reaction Pathway for Metallic Copper Formation

4.1 Two-Stage Mechanism

Combined XRD analysis of quenched samples and DTA-TGA studies have established a **two-stage reaction pathway** for metallic Cu formation in the $Cu(NO_3)_2$-glycine system :

Stage 1 (Combustion Front):

$Cu(NO_3)_2$ decomposes to form Cu_2O and CuO oxide phases through endothermic decomposition. This occurs at early stages in the reaction front.

Stage 2 (Post-Combustion Zone):

The oxide phases are reduced to metallic Cu by reducing gases (H_2, CO, NH_3) generated from fuel decomposition. This reduction occurs in post-combustion zones where temperatures remain elevated and reducing species are abundant .

The necessary condition for SCS of metals is that the oxidizer must decompose with formation of **HNO_3 species**. This decomposition pathway produces NO_2/N_2O species that participate in exothermic reaction with NH_3 from glycine decomposition, sustaining the combustion wave. For compositions with excess glycine, a hydrogen-reducing atmosphere develops in the reaction front .

4.2 Thermodynamic Driving Force for Reduction

The reduction of CuO/Cu_2O to metallic Cu is thermodynamically favorable under the reducing atmosphere generated by excess fuel. The relevant reactions include:

$$
\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}
$$

$$
\text{Cu}_2\text{O} + \text{H}_2 \rightarrow 2\text{Cu} + \text{H}_2\text{O}
$$

$$
\text{CuO} + \text{CO} \rightarrow \text{Cu} + \text{CO}_2
$$

The equilibrium constants for these reactions strongly favor metallic Cu formation at the temperatures prevailing in the post-combustion zone (typically 600-1000 K) when sufficient reducing gas is present .

4.3 Alloy Formation

In Cu-Ni alloy synthesis, a fraction of intermetallic Cu-Ni phase appears **directly in the combustion front**, whereas the remainder forms through reduction of oxide phases in the post-combustion zone . This suggests that the reduction kinetics and pathway depend on the specific metal and alloy composition.

5. Implications for One-Dimensional Nanostructure Formation

5.1 Thermodynamic Constraints

The thermodynamic framework reveals fundamental constraints for 1D metallic nanostructure formation via SCS:

1. **Metallic phase requires fuel-rich conditions ($\varphi > 1$)** to generate reducing atmosphere .

2. **Fuel-rich conditions maximize T_{ad}**, which promotes isotropic nanoparticle formation through enhanced sintering and grain growth.

3. **The two-stage oxide-to-metal reduction pathway** destroys any 1D oxide morphology that might form in the combustion front.

5.2 Kinetic Constraints

Kinetic analysis reveals additional constraints:

1. **SCS reactions complete in seconds to minutes**, far too rapid for the sustained directional growth required for nanowire formation .

2. **The isotropic exothermic driving force** promotes uniform nucleation and growth in all directions.

3. **Rapid gas evolution** disrupts incipient anisotropic structures, favoring porous, agglomerated powders .

5.3 Comparison with Solution-Phase Growth

The solution-phase synthesis of Cu nanowires operates under fundamentally different thermodynamic and kinetic regimes. Meng and Jin demonstrated that Cu NW growth is driven by **axial screw dislocations** along the ⟨110⟩ direction, with associated Eshelby twist . This dislocation-driven growth mechanism provides continuous step edges for anisotropic deposition.

The **reduction-oxidation electrochemical reaction** forming Cu NWs allows precise control of supersaturation through reducing agent concentration, pH, and complexing ligands (EDA, OH^-) . EDA and OH^- serve dual roles: providing suitable pH for redox reaction and complexing Cu^{2+} to lower supersaturation, which promotes dislocation-driven growth .

These conditions—controlled supersaturation, selective capping, and dislocation-driven growth—have no analog in the rapid, high-temperature, isotropic SCS process.

6. Conclusions

Thermodynamic and kinetic analysis of solution combustion synthesis provides fundamental insights into the constraints governing metal nanostructure formation:

1. **Thermodynamics**: T_{ad} increases with φ, reaching values up to ~2480 K. Metallic Cu formation requires $\varphi > 1$ to establish a reducing atmosphere. Gibbs free energy minimization predicts product composition and T_{ad} for any fuel-oxidizer combination.

2. **Kinetics**: Activation energies are lower for fuel-rich (metal-forming) compositions than for fuel-lean (oxide-forming) compositions. The rate-limiting stage shifts from endothermic reactant decomposition to exothermic coordinate compound decomposition with increasing φ.

3. **Reaction pathway**: Metallic Cu forms through a two-stage mechanism: oxide formation in the combustion front followed by reduction in the post-combustion zone. This pathway is incompatible with 1D morphology preservation.

4. **1D growth limitation**: The rapid, isotropic, and high-temperature nature of SCS prevents the sustained anisotropic growth required for nanowire formation. Solution-phase methods employing screw dislocation-driven growth remain the predominant route to Cu nanowires.

These findings establish that SCS is thermodynamically and kinetically unsuitable for direct synthesis of metallic Cu nanorods or nanowires, though it remains an effective technique for CuO nanorods and metallic Cu nanoparticles.

References

[1] Thermodynamics and kinetics of solution combustion synthesis: $Ni(NO_3)_2$ + fuels systems, Combust. Flame 221 (2020) 11–23.

[2] A. Kumar, E.E. Wolf, A.S. Mukasyan, Solution combustion synthesis of metal nanopowders: Copper and copper/nickel alloys, AIChE J. 57 (2011) 3473–3479.

[3] F. Meng, S. Jin, The solution growth of copper nanowires and nanotubes is driven by screw dislocations, Nano Lett. 12 (2012) 234–239.

[5] A.A. Khort, A.B. Tarasov, A.S. Mukasyan, G.V. Trusov, K.B. Podbolotov, S.I. Roslyakov, Solution combustion synthesis of copper nanopowders: The fuel effect, Combust. Sci. Technol. 189 (2017) 1878–1890.

[6] Study of solution combustion synthesis in accelerated rate adiabatic calorimetry for making nickel nanoparticles, Process Saf. Environ. Prot. (2025).

[7] S. Bhanushali, P. Ghosh, A. Ganesh, W. Cheng, 1D copper nanostructures: Progress, challenges and opportunities, Small 11 (2015) 1232–1252.

[9] A. Kumar, E.E. Wolf, A.S. Mukasyan, Solution combustion synthesis of metal nanopowders: Nickel—Reaction pathways, AIChE J. 57 (2011) 2207–2214.

[16] Thermodynamics and kinetics of solution combustion synthesis: $Ni(NO_3)_2$ + fuels systems, OSTI.GOV.

[18] F. Meng, S. Jin, The solution growth of copper nanowires and nanotubes is driven by screw dislocations, Nano Lett. 12 (2012) 234–239 (Supporting Information).